Contact

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www.linkedin.com/in/subhasisdatta (LinkedIn)
www.datasciencecentral.com/profile/SubhasisDatta (Other)
github.com/subhasisdatta (Other)
www.twitter.com/DataPredictive (Personal)

Top Skills

Big Data
Data Science
Predictive Analytics

Certifications

Certified Knowledge Management Professional (CKMP)
A Crash Course in Data Science
Data Science and Analytics Career Paths and Certifications
Apache Hadoop Essentials
Introduction to Big Data

Honors-Awards

Directors Awards and Program Recognition Awards (Several Awards)
Special Performance Recognition Awards (Half dozen Awards)
Corporate Knowledge Management Award
Awards of Excellence and Special Recognition Awards (Over a dozen Awards)
Special Commendation and Service to DoD

Publications

The DATA Act - IT Infrastructure Guidance
“Data Mining”, Critical Review and Technology Assessment (CR/TA) Report

Subhasis Datta MBA, MS, PhD (ABD), Chief Data Scientist

Chief Data Scientist & Machine Learning Officer, AI/ML Expert/ Thought Leader
Washington DC-Baltimore Area

Summary

Successful track record in Transformation, Innovation, Systems and Process improvements, Thought Leadership, Data Science, Big Data, Predictive Analytics, Fraud, Waste and Abuse Detection, Compliance Improvement, Program delivery, Outsourcing, Cost Savings, Systems Engineering and Software development for Federal Civilian and DoD programs as well as COTS. Continued progressive responsibilities and Leadership as Officers of multiple companies providing advice and guidance at the highest levels of Government including Commissioners and Administrators, and General Officers in DoD.

Experienced in Program and Project Management, Strategic Planning, Information Technology, Software and Systems Engineering, Enterprise, Systems and IT Architecture, Business Intelligence, Predictive Modeling, Decision Analytics, Big Data and Data Science as Data Scientist, Modeling and Simulation, Optimization, Process and Systems Modeling, Large Scale Big Data projects, Data Warehousing, Data Mining and Visualization, Technology and Alternatives Assessment.

Significant experience and proven track record with multiple awards in transforming and improving programs and systems, and consulting in Federal Civilian Agencies such as IRS, USDA, HHS, DHS, etc. and in DoD for the US Air Force, Army, Navy, Marines, Secretary of Defense, ODNI, and others.

As Director of Systems & Software Engineering, led large software teams for customized software development in programs at Federal Agencies and DoD services as well as COTS software development in Modeling and Simulation, Transportation Modeling and Logistics

Replication and Extension of Two Leading Agent-Based Tax Compliance Models as a Foundation to Analyze the Tax Ecosystem

Using nonlinear regression procedures for approximate function optimization

Unique Entity Identifiers and the Global Legal Entity Identifier (LEI): A Potential Cornerstone for Transparency of Federal Spending Data and the DATA Act

Optimization, Security and Vulnerability Assessment, Incident and Command Management.

Significant experience in diverse large scale Computing and Data Centers from Mainframes, Unix Servers to Desktops, Networks as Academic, Institutional Research Director, Computer Center Administrator at one of the largest Universities.

Taught as Adjunct Faculty at Robert H. Smith School of Business.

Experience

DMI (Digital Management, LLC)

Chief Data Scientist, Advanced Analytics (AI/ML) Lead, Technology PM, IA/RPA Solution Architect

2020 - Present (6 years)

Vienna, Virginia, United States

As Sr. Data Scientist and ML/AI Expert/Thought Leader responsible for leading Predictive Modeling, application of Machine Learning and AI (ML/AI) for USCIS Chief Data Office (CDO). Explored various ML/AI algorithms in Python and Databricks to address a range of Advanced Analytics needs for CDO. As Advanced Analytics Lead for USCIS CDO will lead a team of Data Scientists, ML/AI, and Data Engineers to address a range of Network, Graph Analytics, NLP, Intelligent Automation using Robotics Process Automation, Risk, and Fraud Advanced Analytics problems. As Technical Lead at SEC providing Technical Program Management support through the lifecycle to a range of Enforcement and Analytics programs across the enterprise on a host of platforms including AWS Cloud, On-Premise.

• Conducted Predictive Modeling using Time Series ARIMA modeling in Python for building MVP models for predicting workload at USCIS; facilitated scaling models several orders of magnitude in Databricks environment with Apache Spark and Netezza
• As Tech Lead conducted Technical Program Management across a portfolio of projects to provide Thought Leadership, solutions in Enterprise Architecture, Systems Lifecycle, Infrastructure, Systems/Software Engineering, Project/ Product Management

Simple Technology Solutions, Inc. an SBA 8(a) & HUBZone Company

CHIEF DATA SCIENTIST, PRINCIPAL, MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE
2019 - 2020 (1 year)
Washington DC-Baltimore Area

Lead STS Enterprise-wide practice in Data Science, Machine Learning and AI across multiple Agencies including DHS (TSA, USCIS, CBP, etc.), HHS, other federal agencies.
Design and build enterprise-wide Data Science, ML, and AI models for risk-based scoring for threat analytics, performance engineering and workload management for TSA
Support Innovation Lab Predictive Modeling for USCIS by envisioning, designing and building innovative solutions for research and technology application using latest tools and techniques in ML/AI

Analytica Inc.

CHIEF DATA SCIENTIST, DIRECTOR OF MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE
2018 - 2019 (1 year)
Washington DC-Baltimore Area

Designed End-to-End Architecture and Prototype for DHS Person-to-Person Matching, Enforcement and Benefits fraud; employing Cloud-based Enterprise data ingest, SAS based Analytics, Big Data Infrastructure, Tableau, and Visual Analytics to identify abuse.
Building ML/AI models for Outlier detection and Performance Analysis with no labeled data for financial fraud/abuse detection
Responsible for identifying financial fraud, maleficence in working with the Federal Financial Oversight Enforcement agency. Build an end-to-end prototype for DHS Enforcement and Benefits fraud, Person-to-Person matching. Advice HHS/CMS on Program Safeguard Fraud and Abuse, SME to IRS in Cloud-based Advanced Analytics, ML/AI applications.

SES Corporation

CHIEF DATA SCIENTIST AND TECHNICAL PROGRAM MANAGER
2017 - 2018 (1 year)
Washington DC-Baltimore Area

Technical Program Manager and Chief Data Scientist for the largest Federal Infrastructure development, Integration and Analytics Program for the 2020 Decennial Census, US Census Bureau.
Responsible for hundreds of Infrastructure Engineers (Cloud, On-Prem, Networking), DevOps and Security Engineers building thousands of Servers

for capturing, managing and analyzing the largest data capture, management, and dissemination of 2020 Census.
Managed Program Schedule, Performance, Budget and Analytics, Metrics.
Advised, designed Cloud-based Enterprise Fraud Analytics with PostgreSQL, SAS, Python/R Analytics on Enterprise Data Lake
Led Infrastructure, DevOps, Security Engineering team with deploying, integrating thousands of cloud servers, program schedule, performance management, metrics analytics on hundreds of millions in Infrastructure operations, hundreds of component ATOs

ICF International

CHIEF DATA SCIENTIST, TECHNICAL DIRECTOR, AND PROGRAM MANAGER

2013 - 2017 (4 years)

Washington DC-Baltimore Area

Transformed as PM/Lead, one of the largest federal IT Infrastructure outsourcing program valued at over $1.5B.
Managed cross-functional teams with Prime contractors and Agency responsible for Data Center, Mainframes, mid-tier Servers, Network Operations, Desktop and Mobile computing assessing their Performance Metrics as guided by their SLAs Other responsibilities included network security monitoring through Trend Micro, event monitoring through Datadog, SSL and DNS management, Docker container builds and deployments.
Automated and improved Incident Management of millions of incident records analytics for root cause analysis, infrastructure downtimes, bottlenecks using data and text mining to identify patterns to improve operations by 15%
Performed full lifecycle end to end program management including task, labor, schedule, cost management and delivery automation resulting in 100% on time delivery of 1,000 deliverables with 10% budget cost savings
Reviewed and analysed Big Data Architecture for USPS postal mail bar coding operations for billions of pieces of mail
Conducted automated performance analytics of 100+ metrics for 12 SLAs and Survey Analytics leading to 10% improvement

MITRE

CHIEF DATA SCIENTIST, ENTERPRISE SYSTEMS ENGINEER AND ARCHITECT

2006 - 2013 (7 years)

Washington DC-Baltimore Area

Transformed as PM, one of the world’s largest ‘Big Data’ fraud detection systems while increasing revenue protection by 2,500% from $1B Year 1 to

$25B+ by Year 4. Collaborated with executive leadership, cross-functional teams in building a new ML/AI fraud prevention program, infrastructure to handle billions of transactions, scoring 200M returns real-time at over 90% accuracy

Increased effectiveness of Technical Program Delivery, Business Understanding by order of magnitude through design and teaching of '6 Day Boot camp' for 100-600 level Mission specific training courses for 300 customers and program team members

Reduced risk in 50 mission-critical systems (multi-billion budget) by 20% by designing a comprehensive risk and readiness assessment framework with multi-dimensional factors and criteria to evaluate operational readiness and address key risks

Improved operational system performance of $ multi-billions by 25% by developing end-to-end IRS 'System of Systems' architectural diagrams, dynamic Simulation Model for C-level Executive what-if analysis scenarios for critical decision-making

Alion Science and Technology

ASSISTANT VICE PRESIDENT, DIRECTOR, COE FOR DATA SCIENCE, MACHINE LEARNING & AI

1996 - 2006 (10 years)

Washington DC-Baltimore Area

Designed analytical and pattern analysis algorithms for one of the largest Insider Threat Unauthorized Access Detection programs mining billions of audit trail transactions for abusive patterns mandated by Congressional law. Build engineering team and scaled algorithms and infrastructure as TPM for delivery within 9 months. The program detected $100'sM in abuse, misdemeanor cases

Designed, Built, Deployed as Lead, $70M Air Force (AF) Predictive Readiness System employing AI, Fuzzy Logic to predict real-time AF Readiness of Forces analyzing billions of metrics and geospatial data utilizing Modelling & Simulation, Visualization

University of Maryland

DIRECTOR OF RESEARCH & INFORMATION SYSTEMS

1990 - 1996 (6 years)

Washington DC-Baltimore Area

Transformed University Financial Management System that handled $30M budget for thousands of students and courses, hundreds of faculties, dozens of departments, colleges and $10M in expenses

Reengineered a manual spreadsheet-based system to automated data-driven system with advanced analytics improving decision-making time by several orders of magnitude, saving millions of dollars. Provided advanced analytical institutional analysis to better understand students

University of Maryland - Robert H. Smith School of Business
ADJUNCT FACULTY AND INSTRUCTOR
1990 - 1991 (1 year)
College Park, MD

Taught various Information Systems, Information Technology and Computer Science courses at Undergraduate level. Facilitated teaching MBA and MS Graduate courses in Information Systems and other Technology disciplines.

Education

Massachusetts Institute of Technology
MicroMasters and Professional Education & Certification Series (expected), Data Science, Artificial Intelligence, Deep Learning, Big Data, IoT and Big Data Analytics · (2015 - 2021)

University of Maryland - Robert H. Smith School of Business
Ph.D (ABD - Advanced to Candidacy), Information Systems, Management Science

University of Maryland - Robert H. Smith School of Business
Master of Science (MS), Information Systems, Management Science

The Johns Hopkins University
Master of Systems Engineering, Systems Engineering (Expected end 2021) · (2021)

Southern Illinois University, Carbondale
Master of Business Administration (MBA), Information Systems, Operations Research